

1. Introduction

Network intrusion detection is an important part of cybersecurity because it helps organization's identify malicious activity on a network. Modern cyberattacks such as brute-force attacks and data exfiltration can compromise sensitive information if they are not detected early.

This project simulated a real-world network intrusion scenario using virtual machines. The project involved configuring a Network Intrusion Detection System (NIDS) using Snort, analyzing packets using Wireshark, performing SSH brute-force attacks using Hydra, and implementing incident response measures.

2. Objectives

The objectives of this project were:

- To configure a virtual network environment.
- To configure static IP addresses.
- To install and configure Snort IDS.
- To monitor and analyze network traffic using Wireshark.
- To simulate SSH brute-force attacks.
- To detect malicious activity using Snort rules.
- To analyze TCP traffic and SSH packets.
- To implement incident response and security hardening.

3. Environment Setup

The virtual environment was created using Oracle VirtualBox.

Two virtual machines were used:

Virtual Machine	Purpose	IP Address
Kali Linux	Attacker Machine	192.168.56.10
Ubuntu VM	Victim + Snort IDS	192.168.56.30

Virtual Machine	Purpose	IP Address
-----------------	---------	------------

Both virtual machines used:

- Adapter 1: Bridged (Internet Access)
- Adapter 2: Internal Network (cyberlab)

Promiscuous mode was set to “Allow All”.

Static IP Configuration

Static IP addresses were configured using Netplan on Ubuntu and NetworkManager on Kali Linux.

Command used:

`sudo nano /etc/netplan/00-installer-config.yaml`

`sudo netplan apply & nmtui for kali`

Ubuntu

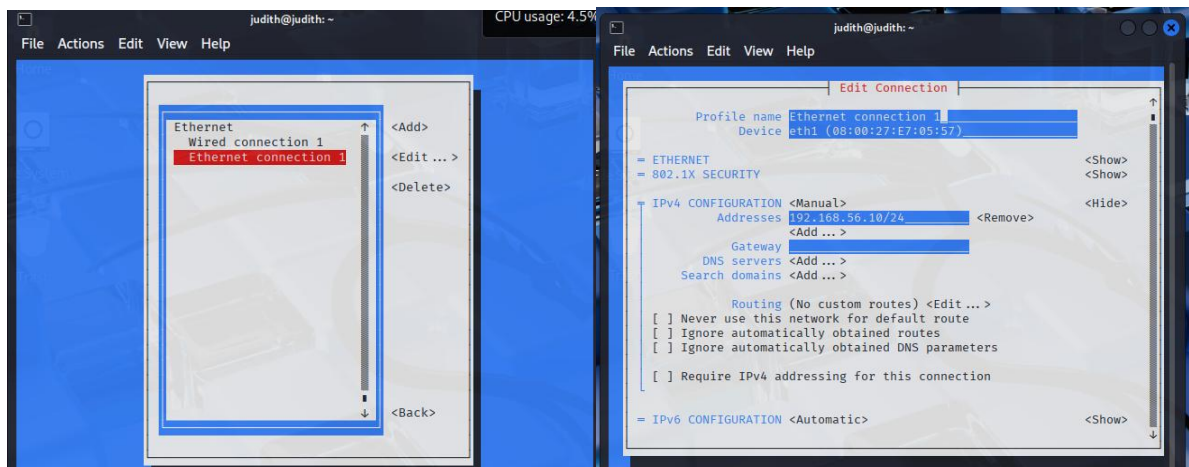
The screenshot shows a terminal window titled 'judith@ids-vm: ~'. The user is editing the file '/etc/netplan/00-installer-config.yaml' using the GNU nano 6.2 editor. The configuration file content is as follows:

```
network:
  version: 2
  ethernet:
    enp0s3:
      dhcp4: true

    enp0s8:
      dhcp4: false
      addresses:
        - 192.168.56.30/24
```

The terminal window also displays a help menu at the bottom with various keyboard shortcuts for nano editor operations.

Kali



4. Connectivity Testing

Connectivity between the attacker and victim machines was verified using ICMP ping requests.

Commands used:

```
ping 192.168.56.30
```

```
ping 192.168.56.10
```

Successful replies confirmed communication between the virtual machines.

```
(judith@judith)-[~]
$ ping 192.168.56.30
PING 192.168.56.30 (192.168.56.30) 56(84) bytes of data.
64 bytes from 192.168.56.30: icmp_seq=1 ttl=64 time=1.02 ms
64 bytes from 192.168.56.30: icmp_seq=2 ttl=64 time=1.11 ms
64 bytes from 192.168.56.30: icmp_seq=3 ttl=64 time=1.34 ms
^Z
zsh: suspended ping 192.168.56.30
(judith@judith)-[~]
$
```

```
judith@ids-vm:~$ ping 192.168.56.10
PING 192.168.56.10 (192.168.56.10) 56(84) bytes of data.
64 bytes from 192.168.56.10: icmp_seq=1 ttl=64 time=0.819 ms
64 bytes from 192.168.56.10: icmp_seq=2 ttl=64 time=0.893 ms
64 bytes from 192.168.56.10: icmp_seq=3 ttl=64 time=1.20 ms
64 bytes from 192.168.56.10: icmp_seq=4 ttl=64 time=1.01 ms
64 bytes from 192.168.56.10: icmp_seq=5 ttl=64 time=1.87 ms
^Z
[3]+  Stopped                  ping 192.168.56.10
judith@ids-vm:~$
```

5. SSH Configuration

SSH was installed and enabled on the Ubuntu victim machine to allow remote access simulation.

Commands used:

`sudo apt update`

`sudo apt install openssh-server -y`

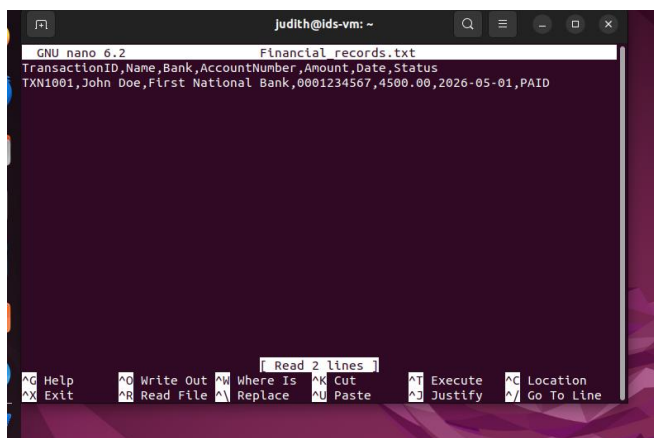
`sudo systemctl status ssh`

```
judith@judith-VirtualBox:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2026-05-26 13:41:03 CAT; 1min 44s ago
     Docs: man:sshd(8)
           man:sshd_config(5)
   Main PID: 2807 (sshd)
    Tasks: 1 (limit: 2267)
   Memory: 1.7M
      CPU: 32ms
   CGroup: /system.slice/ssh.service
           └─2807 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"
```

A fake sensitive file named `financial_records.txt` was also created to simulate confidential organisational data.

Command used:

`nano financial_records.txt`



```
GNU nano 6.2 Financial records.txt
TransactionID,Name,Bank,AccountNumber,Amount,Date,Status
TXN1001,John Doe,First National Bank,0001234567,4500.00,2026-05-01,PAID
```

6. Snort Installation and Configuration

Snort IDS was installed on the Ubuntu VM to monitor and detect malicious traffic.

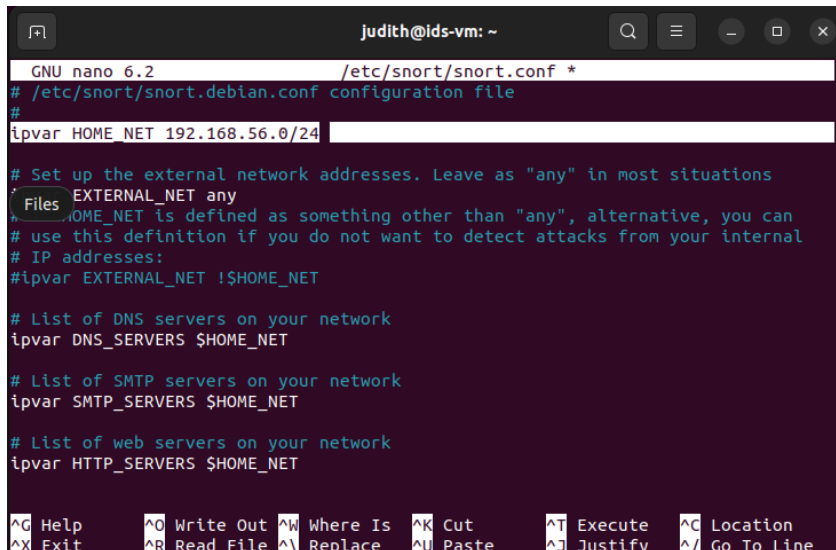
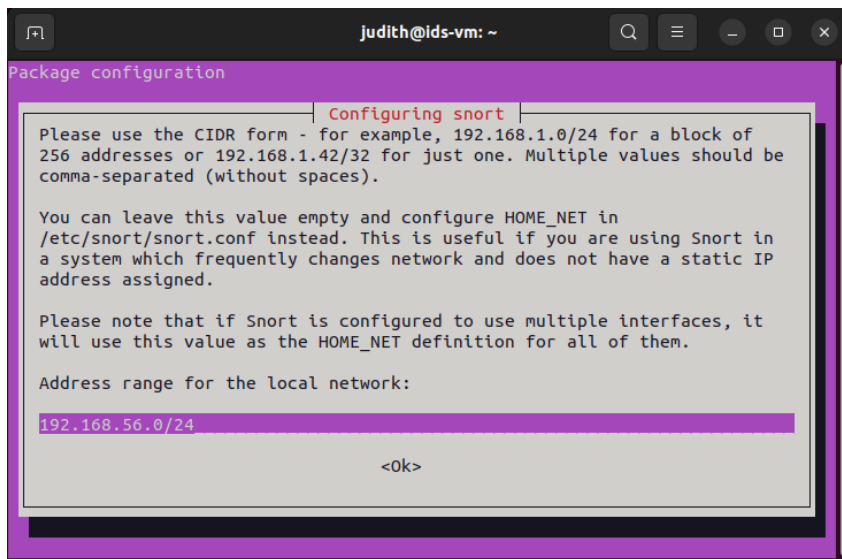
Commands used:

`sudo apt update`

`sudo apt install snort -y`

During installation, the HOME_NET was configured as:

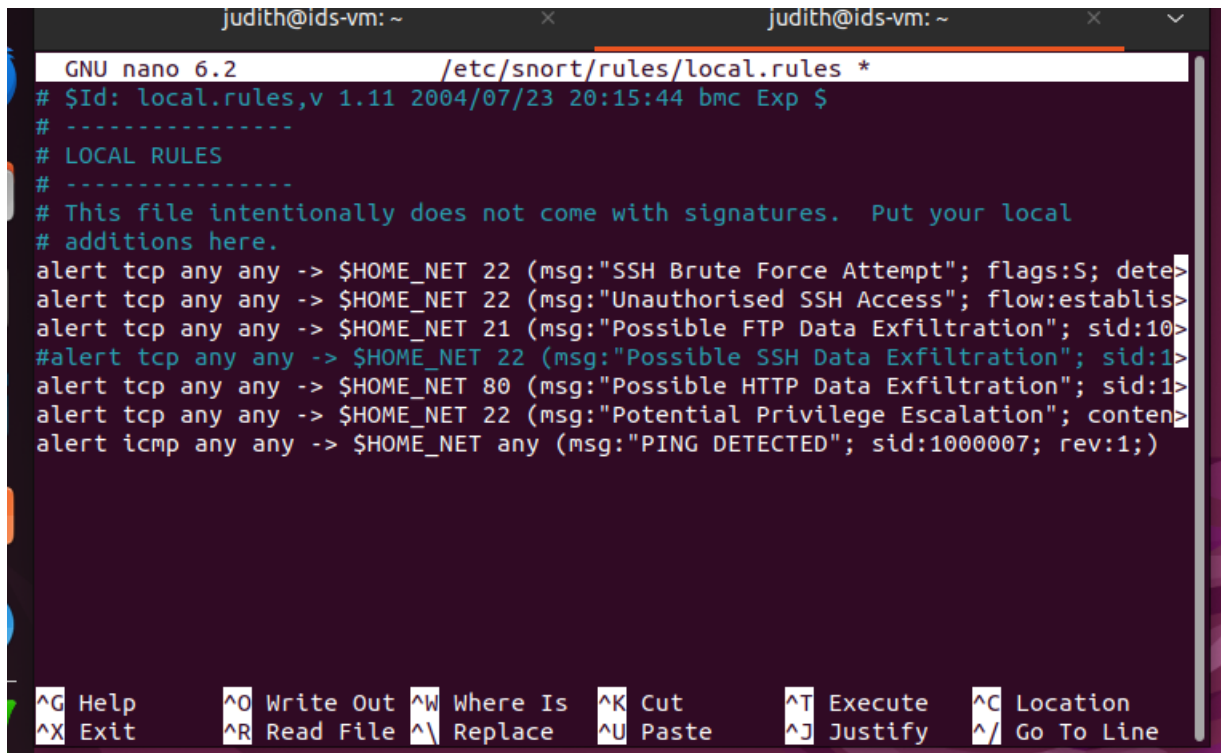
192.168.56.0/24



7. Custom Snort Rules

Custom rules were created inside:

sudo nano /etc/snort/rules/local.rules



```
GNU nano 6.2 /etc/snort/rules/local.rules *
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
# -----
# LOCAL RULES
# -----
# This file intentionally does not come with signatures.  Put your local
# additions here.
alert tcp any any -> $HOME_NET 22 (msg:"SSH Brute Force Attempt"; flags:S; dete>
alert tcp any any -> $HOME_NET 22 (msg:"Unauthorised SSH Access"; flow:establis>
alert tcp any any -> $HOME_NET 21 (msg:"Possible FTP Data Exfiltration"; sid:10>
#alert tcp any any -> $HOME_NET 22 (msg:"Possible SSH Data Exfiltration"; sid:1>
alert tcp any any -> $HOME_NET 80 (msg:"Possible HTTP Data Exfiltration"; sid:1>
alert tcp any any -> $HOME_NET 22 (msg:"Potential Privilege Escalation"; conten>
alert icmp any any -> $HOME_NET any (msg:"PING DETECTED"; sid:1000007; rev:1;)>

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute  ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify  ^_ Go To Line
```

ICMP Detection Rule

```
alert icmp any any -> $HOME_NET any (msg:"PING DETECTED"; sid:1000007; rev:1;)
```

SSH Data Exfiltration Rule

```
alert tcp $HOME_NET any -> any 22 (msg:"Possible SSH Data Exfiltration"; sid:1000004; rev:1;)
```

SSH Brute Force Rule

```
alert tcp any any -> $HOME_NET 22 (msg:"SSH Brute Force Attempt"; flags:S; threshold:type
both, track by_src, count 5, seconds 60; sid:1000001; rev:1;)
```

Snort configuration was validated using:

```
sudo snort -T -c /etc/snort/snort.conf
```

```
judith@ids-vm: ~  
Using PCRE version: 8.39 2016-06-14  
Using ZLIB version: 1.2.11  
Rules Engine: SF_SNORT_DETECTION_ENGINE Version 3.1 <Build 1>  
ld 1>  
Preprocessor Object: SF_DCERPC2 Version 1.0 <Build 3>  
Preprocessor Object: SF_SDF Version 1.1 <Build 1>  
Preprocessor Object: SF_MODBUS Version 1.1 <Build 1>  
Preprocessor Object: SF_DNS Version 1.1 <Build 4>  
Preprocessor Object: SF_REPUTATION Version 1.1 <Build 1>  
Preprocessor Object: SF_IMAP Version 1.0 <Build 1>  
Preprocessor Object: SF_SSLPP Version 1.1 <Build 4>  
Preprocessor Object: SF_GTP Version 1.1 <Build 1>  
Preprocessor Object: SF_POP Version 1.0 <Build 1>  
Preprocessor Object: SF_DNP3 Version 1.1 <Build 1>  
Preprocessor Object: SF_SSH Version 1.1 <Build 3>  
Preprocessor Object: SF_SMTP Version 1.1 <Build 9>  
Preprocessor Object: SF_SIP Version 1.1 <Build 1>  
Preprocessor Object: appid Version 1.1 <Build 5>  
Preprocessor Object: SF_FTPTELNET Version 1.2 <Build 13>  
Snort successfully validated the configuration!  
Snort exiting  
judith@ids-vm:~$
```

Snort was started using:

```
sudo snort -A console -q -u snort -g snort -c /etc/snort/snort.conf -i enp0s8 -b
```

```
judith@ids-vm:~$ sudo snort -A console -q -u snort -c /etc/snort/snort.conf -i enp0s8 -b
```

8. Attack Simulation

Hydra was used on Kali Linux to simulate an SSH brute-force attack against the Ubuntu machine.

A password list was created:

```
nano passwords.txt
```

```
judith@judith: ~  
File Actions Edit View Help  
GNU nano 8.4 passwords.txt *  
123456  
password  
kali  
student  
admin
```

Hydra command used:

```
hydra -l judith -P passwords.txt ssh 192.168.56.30
```

Brute force attack from Kali

```
(judith@judith)-[~]  
$ hydra -l judith -P passwords.txt ssh://192.168.56.30/  
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in  
military or secret service organizations, or for illegal purposes (this is n  
on-binding, these ** ignore laws and ethics anyway).  
  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-05-28 15:  
23:11  
[WARNING] Many SSH configurations limit the number of parallel tasks, it is r  
ecommended to reduce the tasks: use -t 4  
[DATA] max 5 tasks per 1 server, overall 5 tasks, 5 login tries (l:1/p:5), ~1  
try per task  
[DATA] attacking ssh://192.168.56.30:22/  
1 of 1 target completed, 0 valid password found  
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-05-28 15:  
23:15
```

Snort detection

```
judith@ids-vm: ~  
05/28-17:28:40.656447  [**] [1:1000007:1] PING DETECTED [**] [Priority: 0] {ICMP  
} 192.168.56.10 -> 192.168.56.30  
05/28-17:28:40.656489  [**] [1:1000007:1] PING DETECTED [**] [Priority: 0] {ICMP  
} 192.168.56.30 -> 192.168.56.10  
05/28-17:28:41.657691  [**] [1:1000007:1] PING DETECTED [**] [Priority: 0] {ICMP  
} 192.168.56.10 -> 192.168.56.30  
05/28-17:28:41.657753  [**] [1:1000007:1] PING DETECTED [**] [Priority: 0] {ICMP  
} 192.168.56.30 -> 192.168.56.10  
Rhythmbox :55.958317  [**] [1:1000002:1] Unauthorised SSH Access [**] [Priority  
: 0] {TCP} 192.168.56.10:40980 -> 192.168.56.30:22  
05/28-17:29:56.033579  [**] [1:1000002:1] Unauthorised SSH Access [**] [Priority  
: 0] {TCP} 192.168.56.10:40980 -> 192.168.56.30:22  
05/28-17:29:56.338522  [**] [1:1000001:1] SSH Brute Force Attempt [**] [Priority  
: 0] {TCP} 192.168.56.10:41026 -> 192.168.56.30:22  
05/28-17:29:56.338522  [**] [1:1000002:1] Unauthorised SSH Access [**] [Priority  
: 0] {TCP} 192.168.56.10:40990 -> 192.168.56.30:22  
05/28-17:29:56.347352  [**] [1:1000002:1] Unauthorised SSH Access [**] [Priority  
: 0] {TCP} 192.168.56.10:41002 -> 192.168.56.30:22  
05/28-17:29:56.347352  [**] [1:1000002:1] Unauthorised SSH Access [**] [Priority  
: 0] {TCP} 192.168.56.10:40992 -> 192.168.56.30:22  
05/28-17:29:56.347352  [**] [1:1000002:1] Unauthorised SSH Access [**] [Priority  
: 0] {TCP} 192.168.56.10:41010 -> 192.168.56.30:22  
05/28-17:29:56.430946  [**] [1:1000002:1] Unauthorised SSH Access [**] [Priority  
: 0] {TCP} 192.168.56.10:40992 -> 192.168.56.30:22
```

Attack when password is in password file


```
(judith@judith)-[~]
$ hydra -l judith -P passwords.txt ssh://192.168.56.30/
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in
military or secret service organizations, or for illegal purposes (this is n
on-binding, these *** ignore laws and ethics anyway).

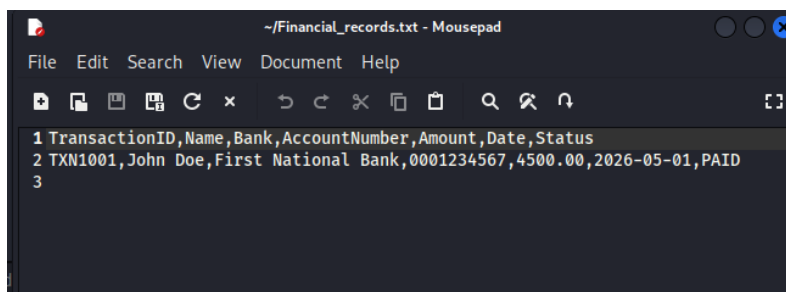
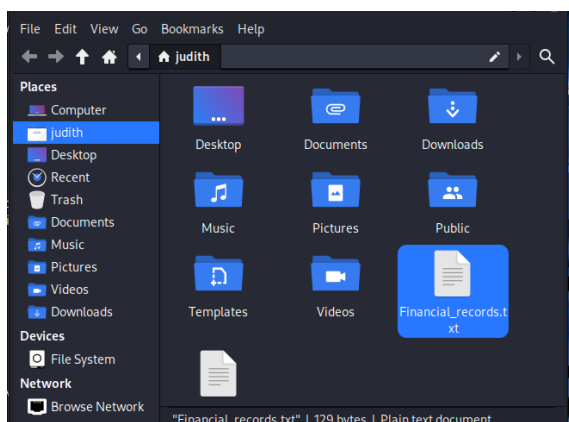
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-05-28 17:
34:52
[WARNING] Many SSH configurations limit the number of parallel tasks, it is r
ecommended to reduce the tasks: use -t 4
[DATA] max 6 tasks per 1 server, overall 6 tasks, 6 login tries (l:1/p:6), ~1
try per task
[DATA] attacking ssh://192.168.56.30:22/
[22][ssh] host: 192.168.56.30 login: judith password: Tuwilika123!
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-05-28 17:
34:55

(judith@judith)-[~]
$
```

```
(judith@judith)-[~]
$ scp judith@192.168.56.30:/home/judith/Financial_records.txt .
judith@192.168.56.30's password:
Financial_records.txt 100% 129 18.4KB/s 00:00

(judith@judith)-[~]
$
```

Attacker gains access to financial records and installs them



9. Wireshark Installation and Packet Analysis

Wireshark was installed to capture and analyse packets.

Commands used:

```
sudo apt install wireshark -y
```

Wireshark was used to inspect SSH traffic and verify TCP communication between the attacker and victim machines.

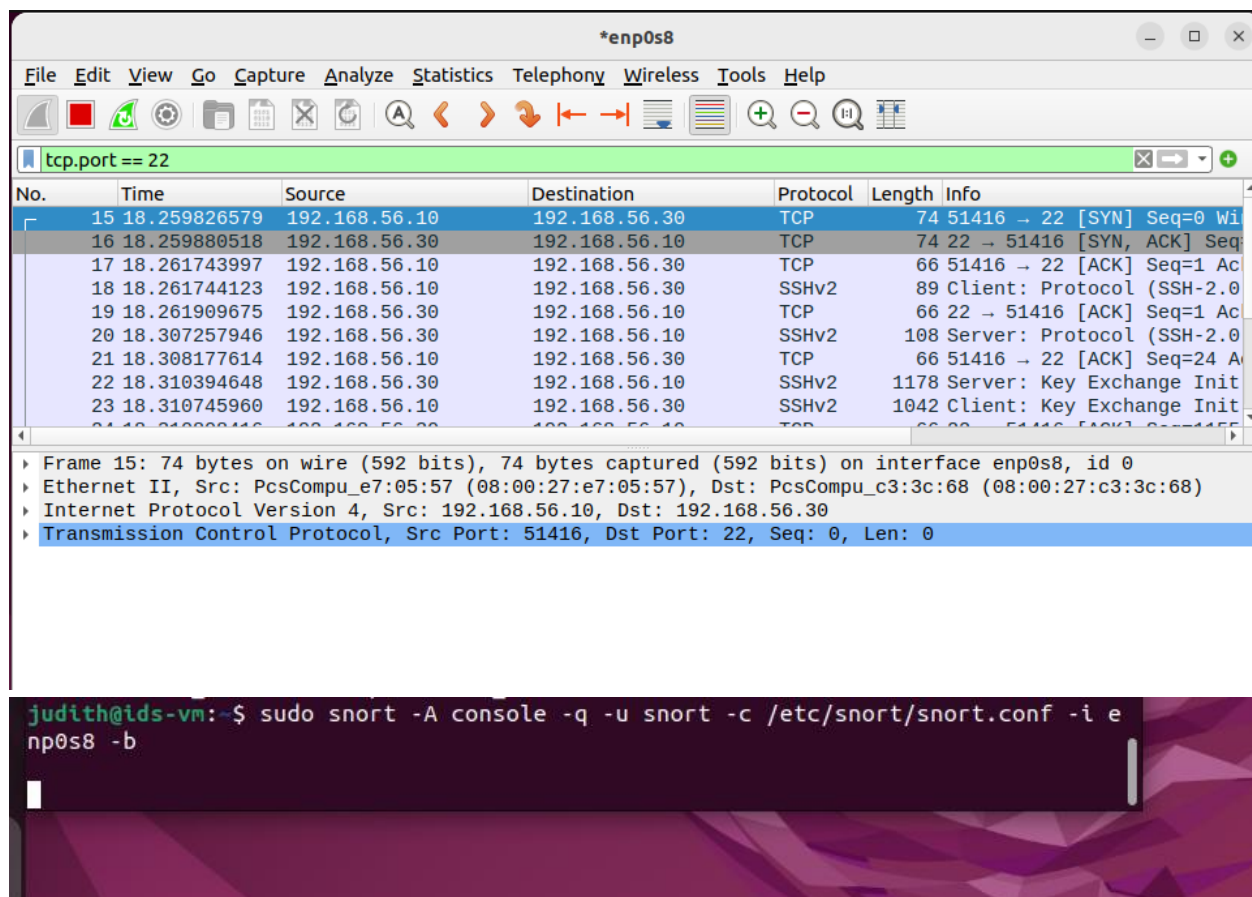
Filter used:

```
tcp.port == 22
```

The packet capture showed:

- SYN packets
- SYN-ACK packets
- ACK packets
- SSHv2 negotiation
- Key exchange initialization

This confirmed a successful TCP 3-way handshake and SSH communication.



10. Detection Results

Snort successfully generated alerts during:

- ICMP ping requests
- SSH traffic
- SSH brute-force attempts
- Data exfiltration

The alerts were displayed in real time on the console.

Wireshark packet analysis captured the SSH traffic generated during the brute-force attack and file transfer process. Snort simultaneously generated alerts identifying suspicious SSH traffic and possible data exfiltration activity.

```

judith@ids-vm:~$ sudo snort -A console -q -u snort -g snort -c /etc/snort/snort.conf -i enp0s8 -b
05/28-15:15:12.551491  [**] [1:1000004:1] Possible SSH Data Exfiltration [**] [P
riority: 0] {TCP} 192.168.56.10:54158 -> 192.168.56.30:22
05/28-15:15:12.554655  [**] [1:1000004:1] Possible SSH Data Exfiltration [**] [P
riority: 0] {TCP} 192.168.56.10:54158 -> 192.168.56.30:22
05/28-15:15:12.561401  [**] [1:1000004:1] Possible SSH Data Exfiltration [**] [P
riority: 0] {TCP} 192.168.56.10:54158 -> 192.168.56.30:22
05/28-15:15:12.689173  [**] [1:1000004:1] Possible SSH Data Exfiltration [**] [P
riority: 0] {TCP} 192.168.56.10:54158 -> 192.168.56.30:22
05/28-15:15:12.736783  [**] [1:1000004:1] Possible SSH Data Exfiltration [**] [P
riority: 0] {TCP} 192.168.56.10:54158 -> 192.168.56.30:22
05/28-15:15:12.755004  [**] [1:1000004:1] Possible SSH Data Exfiltration [**] [P
riority: 0] {TCP} 192.168.56.10:54158 -> 192.168.56.30:22
05/28-15:16:11.840586  [**] [1:1000004:1] Possible SSH Data Exfiltration [**] [P
riority: 0] {TCP} 192.168.56.10:54158 -> 192.168.56.30:22
05/28-15:16:11.883510  [**] [1:1000004:1] Possible SSH Data Exfiltration [**] [P
riority: 0] {TCP} 192.168.56.10:54158 -> 192.168.56.30:22
05/28-15:16:11.892883  [**] [1:1000004:1] Possible SSH Data Exfiltration [**] [P
riority: 0] {TCP} 192.168.56.10:54158 -> 192.168.56.30:22
05/28-15:16:11.948442  [**] [1:1000004:1] Possible SSH Data Exfiltration [**] [P

```

No.	Time	Source	Destination	Protocol	Length	Info
15	18.259826579	192.168.56.10	192.168.56.30	TCP	74	51416 → 22 [SYN] Seq=0 Win=0 Len=0
16	18.259880518	192.168.56.30	192.168.56.10	TCP	74	22 → 51416 [SYN, ACK] Seq=1042 Win=0 Len=0
17	18.261743997	192.168.56.10	192.168.56.30	TCP	66	51416 → 22 [ACK] Seq=1042 Win=0 Len=0
18	18.261744123	192.168.56.10	192.168.56.30	SSHv2	89	Client: Protocol (SSH-2.0)
19	18.261909675	192.168.56.30	192.168.56.10	TCP	66	22 → 51416 [ACK] Seq=1042 Win=0 Len=0
20	18.307257946	192.168.56.30	192.168.56.10	SSHv2	108	Server: Protocol (SSH-2.0)
21	18.308177614	192.168.56.10	192.168.56.30	TCP	66	51416 → 22 [ACK] Seq=2440 Win=0 Len=0
22	18.310394648	192.168.56.30	192.168.56.10	SSHv2	1178	Server: Key Exchange Init
23	18.310745960	192.168.56.10	192.168.56.30	SSHv2	1042	Client: Key Exchange Init

Frame 15: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface enp0s8, id 0
 Ethernet II, Src: PcsCompu_e7:05:57 (08:00:27:e7:05:57), Dst: PcsCompu_c3:3c:68 (08:00:27:c3:3c:68)
 Internet Protocol Version 4, Src: 192.168.56.10, Dst: 192.168.56.30
 Transmission Control Protocol, Src Port: 51416, Dst Port: 22, Seq: 0, Len: 0

11. Incident Response

To reduce future attacks, security hardening measures were considered, including:

- Using stronger passwords
- Restricting SSH access
- Blocking suspicious IP addresses
- Monitoring suspicious traffic continuously
- Applying firewall rules using iptables

Example firewall command:

```
sudo iptables -A INPUT -s 192.168.56.10 -j DROP
```

12. Challenges Encountered

Several challenges were encountered during the project, including:

- YAML indentation errors during Netplan configuration
- Snort rule syntax errors
- VirtualBox packet visibility limitations
- Interface configuration issues
- Wireshark permission issues

These issues were resolved through troubleshooting and configuration adjustments.

13. Conclusion

The project successfully demonstrated a simulated network intrusion detection and response environment. Snort IDS successfully detected suspicious activity such as SSH brute-force attempts and possible data exfiltration. Wireshark provided packet-level visibility into network traffic and confirmed TCP communication between the attacker and victim systems.

The project demonstrated the importance of intrusion detection systems in protecting organisational networks from malicious activity.

14. References

- Snort Documentation
- Wireshark Documentation
- Kali Linux Documentation
- Ubuntu Documentation
- Oracle VirtualBox Documentation